

Course Work

Venue Preference Prediction Using Data Models

Survey Design

Write the report

• Background and Objectives

The coursework analyzes students' venue preferences and predicts them using linear algebra and machine learning. By identifying key factors through data analysis, we aim to improve school space planning and resource allocation.

• Research Methodology

1. Data Collection

Tool:Google Forms

A survey of 30 students collected feedback on six venues , measuring distance (1-9), comfort level (1-9), usage frequency (0-7 days/week), expected venue size (1-9), and overall preference (1-9).

2. Data Analysis & Modeling

We applied **the Least Squares Method** to predict venue preferences by converting variables into a numerical matrix and fitting linear, quadratic, and cubic functions. Error analysis showed the quadratic function had the lowest error, making it the final model. To validate accuracy, we used **Simple ML for Sheets**, and the results closely matched, confirming the model's feasibility.

$$A^T Ax = A^T b$$

3. Application Development

Developed in **MATLAB**, the application allows users to input distance, comfort level, usage frequency, and expected venue size to predict preference scores using the established model. A visual interface displays results for intuitive interpretation.

• Research Findings

- Preferences
- Comfort Impact
- Model Accuracy

This study confirms that linear algebra and machine learning effectively predict venue preferences, aiding space planning.

